

ABSTRACT

This project explores customer segmentation using unsupervised machine learning techniques, focusing on the Wholesale Customers dataset. The primary objective is to identify distinct customer groups based on their annual spending across various product categories such as fresh products, milk, grocery, frozen items, detergents, and delicatessen. Through data preprocessing, dimensionality reduction using PCA, and clustering via KMeans, the project uncovers hidden patterns in consumer behavior. The clustering results provide meaningful insights for targeted marketing strategies and operational decision-making in the wholesale and retail sectors.

INTRODUCTION

In today's competitive retail landscape, understanding customer behavior is essential for strategic decision-making. This project aims to perform unsupervised machine learning to segment wholesale customers based on their purchasing patterns. By analyzing customer data without predefined labels, we can uncover hidden groupings that reflect underlying similarities in buying behavior.

Understanding customer behavior is essential for businesses to tailor services, improve customer satisfaction, and drive sales growth. Traditional approaches to market segmentation often rely on domain knowledge and predefined criteria. In contrast, unsupervised machine learning enables data-driven discovery of patterns without prior labeling. In this project, we analyze the Wholesale Customers dataset using techniques such as Principal Component Analysis (PCA) and KMeans clustering to segment customers based on their purchasing behavior. The aim is to group customers with similar spending profiles, helping businesses to optimize marketing, inventory management, and service personalization.



CONCLUSION

This project aimed to segment wholesale customers based on their annual spending across different product categories using unsupervised machine learning techniques. By applying algorithms such as KMeans, DBSCAN, and Spectral Clustering, we were able to discover natural groupings within the data that are not immediately visible through traditional analysis.

This project successfully demonstrated how unsupervised learning techniques can be applied to segment customers in the Wholesale Customers dataset. By employing PCA for dimensionality reduction and KMeans for clustering, we identified meaningful customer segments based on their purchasing habits. These insights reveal distinct behavioral patterns, such as high spenders on detergents or fresh produce, which can be instrumental for targeted marketing and inventory planning. Overall, unsupervised learning proves to be a powerful tool for extracting actionable intelligence from unlabelled data, providing a foundation for more strategic business decisions.